



OctuKheir
Academy

Pharmaceutical Organic Chemistry II

Content:

Alcohol

- Nomenclature
- Reactions of alcohols
- Reactions of alcohols

For courses & materials:

 01067043749

 @OctuKheir



Lecture: 1

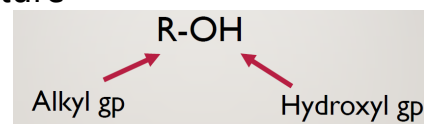
Pages: 6



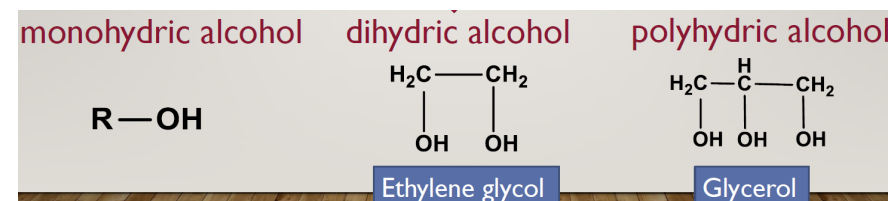
ALCOHOLS

Classification of alcohols

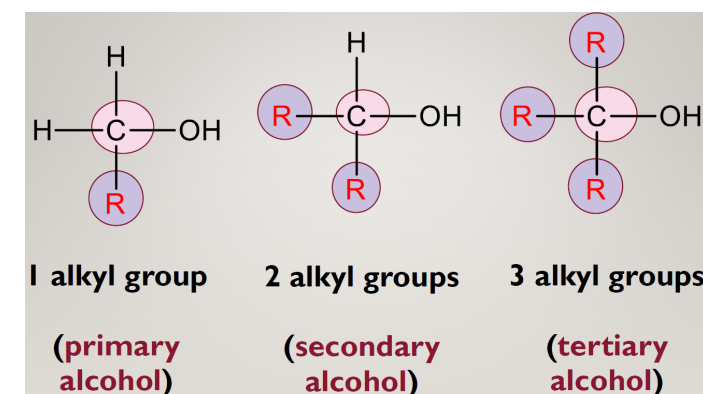
Structure



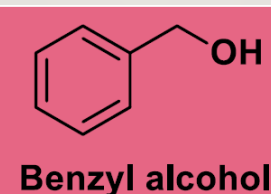
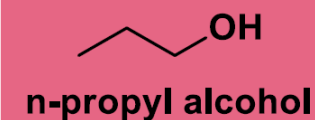
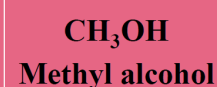
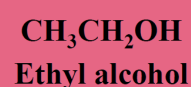
2) According to the number of hydroxyl groups



1) According to the number of alkyl groups attached to the α -carbon

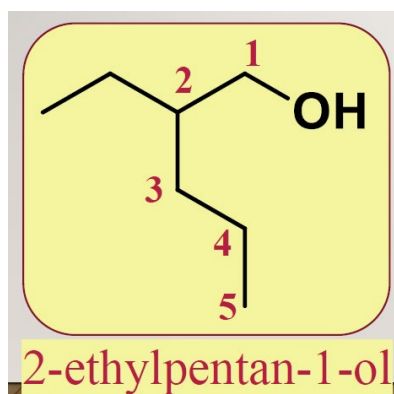


Nomenclature of alcohols: Common name: • Monohydric alcohols: named as alkyl alcohol

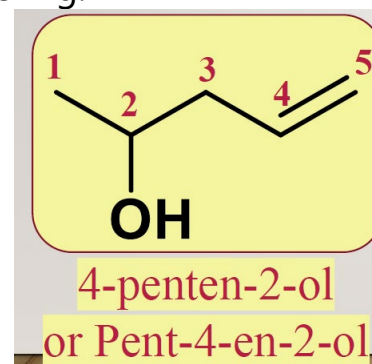


IUPAC name:

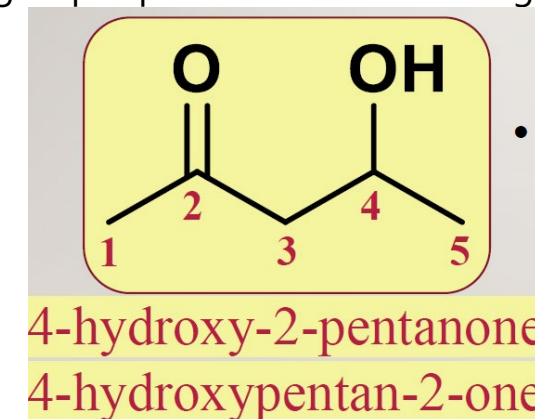
- Select the longest continuous carbon chain which contains hydroxyl group.
- the carbon atom bearing the hydroxyl group take the lowest possible number.



- OH group is preferred over the double or triple bond in numbering.



- C=O group is preferred over the OH group.





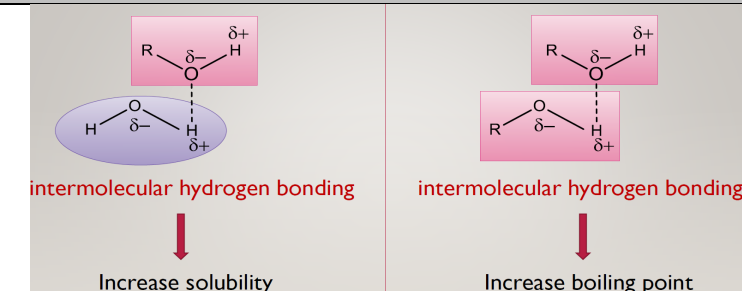
Physical Properties of alcohols

Miscible with water

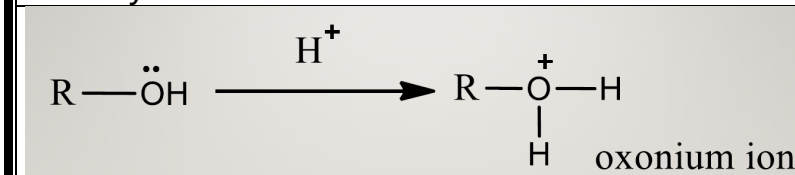
- Due to intermolecular hydrogen bonding between alcohol molecules and water molecules
- As the number of carbon atoms increase, the solubility decrease

High boiling point

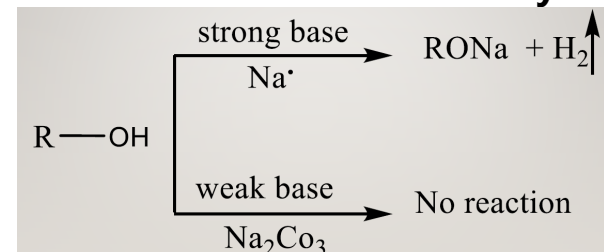
- Due to intermolecular hydrogen bonding between alcohol molecules.
- alcohols have a higher boiling point than the corresponding alkane.



Basicity of alcohols: Alcohols act as weak bases



Acidity of alcohols: Alcohols act as very weak acids.



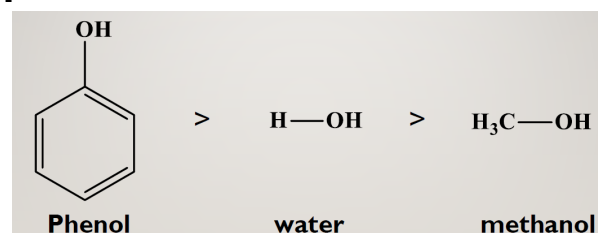
Sodium bicarbonate test:

- gives effervescence with strong acids.
- But no effervescence with weak acids like alcohols (no reaction).

Sodium Bicarbonate Test

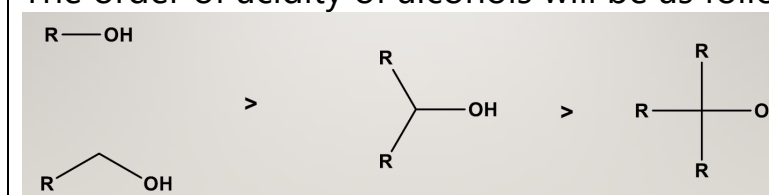


ORDER OF ACIDITY



alkyl groups have +I effect

The order of acidity of alcohols will be as follows: $1^\circ > 2^\circ > 3^\circ$



Preparation of alcohols:

From alkenes

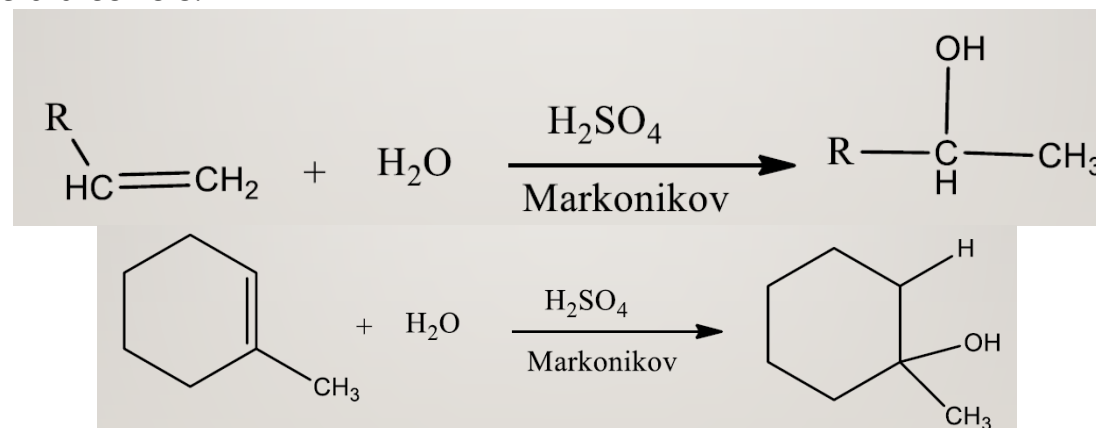
From alkyl halide

From carbonyl compounds

Preparation of alcohols

1) From Alkenes:

When treating alkenes with concentrated sulfuric acid form alkyl hydrogen sulfates which on subsequent hydrolysis with water or steam yield alcohols.



2) From alkyl halide (Nucleophilic substitution reaction):





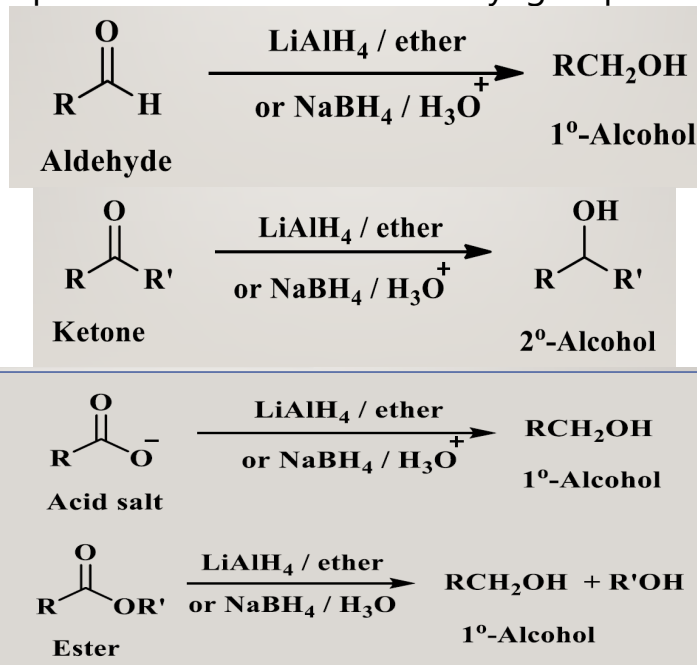
3) From carbonyl compounds

a) Reduction

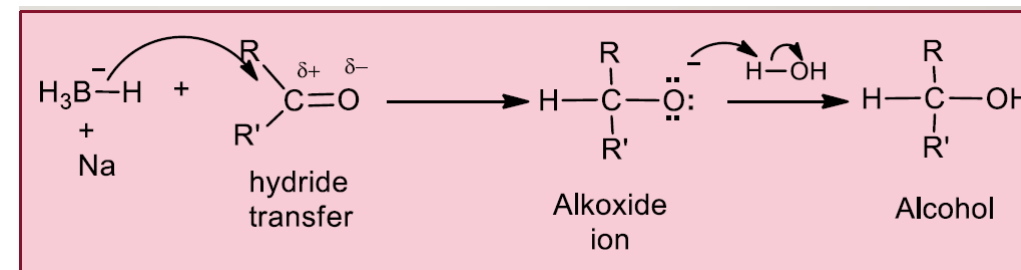
b) Grignard reagent

a) Reduction

Primary and secondary alcohols can be synthesized by the reduction of a variety of compounds that contain carbonyl group.

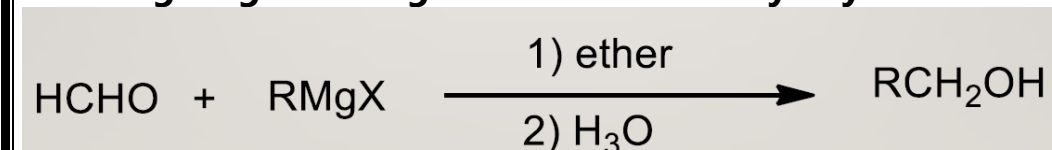


The key step in the reduction of a carbonyl compound by either lithium aluminium hydride or sodium borohydride is the transfer of a hydride ion from the metal to the carbonyl carbon and act as a nucleophile.

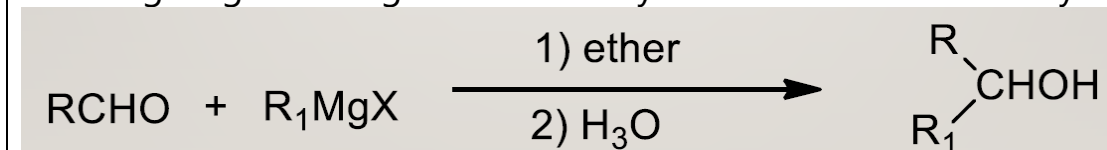


b) Reaction of Grignard reagent with carbonyl compounds

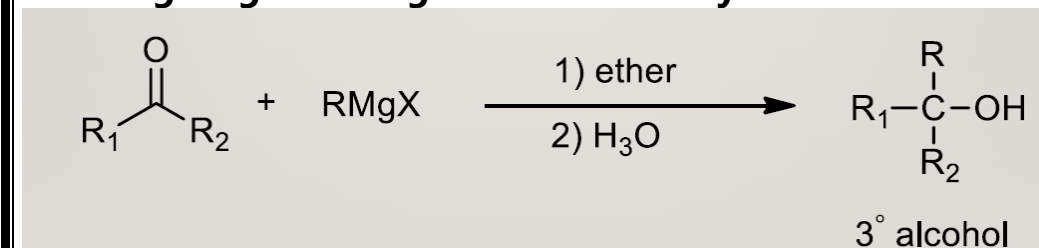
Reacting Grignard reagent with formaldehyde yields 1° alcohols



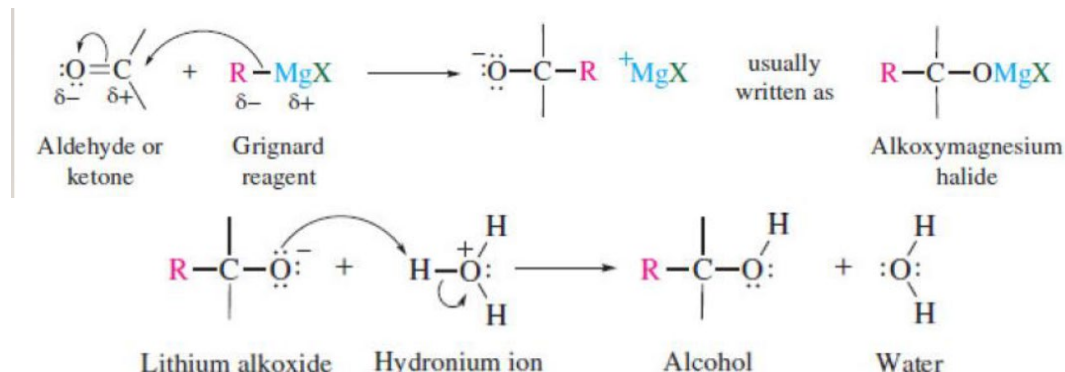
Reacting Grignard reagent with aldehyde other than formaldehyde yields 2° alcohols



Reacting Grignard reagent with ketone yields 3° alcohols



Mechanism:





Reactions of alcohols

1) Nucleophilic substitution reaction (Conversion of alcohol into alkyl halide)

Using hydrogen halide (HX)

Using phosphorus tribromide (PBr₃)

Using thionyl chloride (SOCl₂)

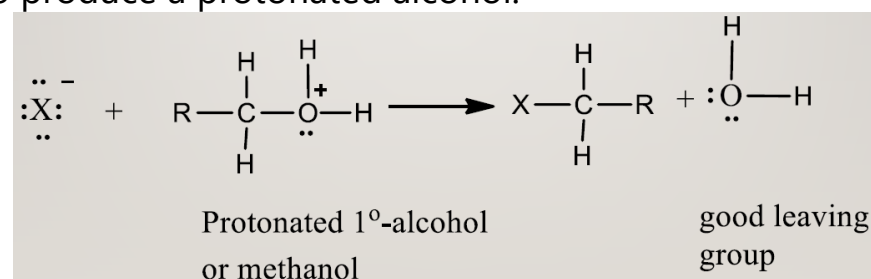
Reaction with ZnCl₂ (Lucas test)

Using hydrogen halide (HX)

The order of reactivity of the hydrogen halide is $\text{HI} > \text{HBr} > \text{HCl}$ (HF is unreactive) while the order of reactivity of alcohols is $3^\circ > 2^\circ > 1^\circ > \text{methyl}$.

The reaction is acid catalyzed and involves the formation of carbocation (hence, rearrangement) in case of secondary, tertiary, allylic and benzylic alcohols.

primary alcohols and methanol react through a mechanism without formation of carbocation (S_N2). In these reactions, The function of the acid is to produce a protonated alcohol.



Mechanism 5.3

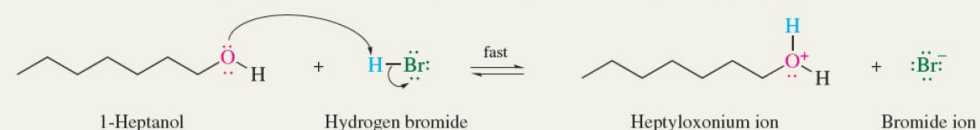
Formation of 1-Bromoheptane from 1-Heptanol and Hydrogen Bromide

THE OVERALL REACTION:

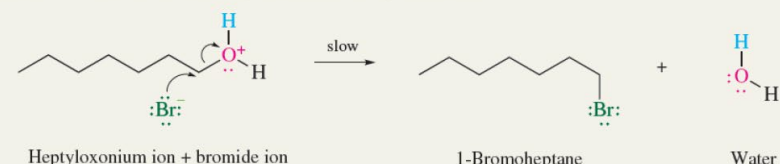


THE MECHANISM:

Step 1: Proton transfer from hydrogen bromide to 1-heptanol to give the corresponding alkyloxonium ion:

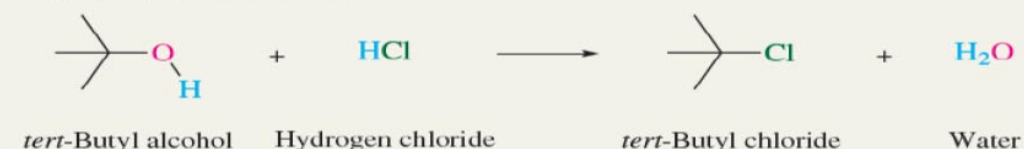


Step 2: Displacement of water from the alkyloxonium ion by bromide:



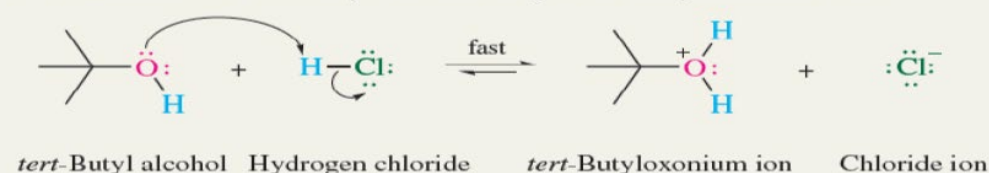
Formation of *tert*-Butyl Chloride from *tert*-Butyl Alcohol and Hydrogen Chloride

THE OVERALL REACTION:

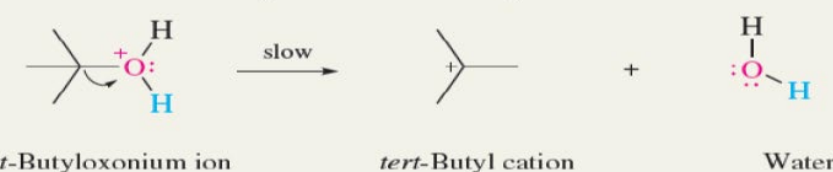


THE MECHANISM:

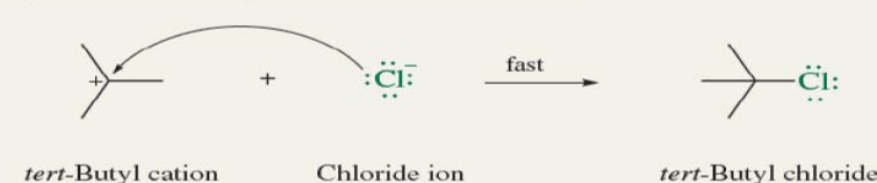
Step 1: Protonation of *tert*-butyl alcohol to give an alkyloxonium ion:



Step 2: Dissociation of *tert*-butyloxonium ion to give a carbocation:



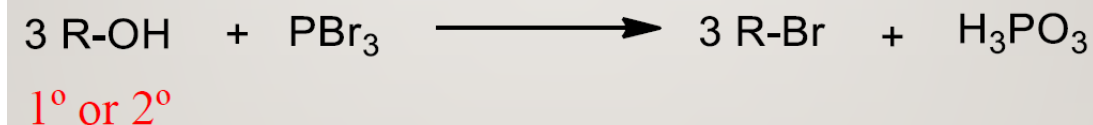
Step 3: Capture of *tert*-butyl cation by chloride ion:





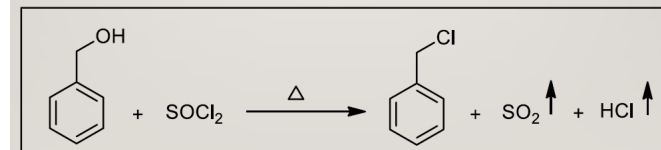
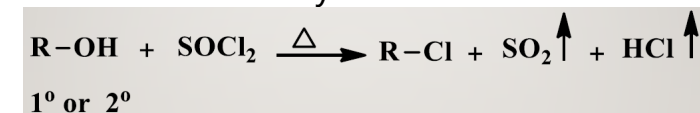
Using phosphorus tribromide (PBr₃)

Unlike the reaction with HBr, the reaction of alcohol with PBr₃, PCl₃ or PCl₅ doesn't involve the formation of carbocation and occurs without rearrangement, consequently, it is preferred than hydrogen halide.



Using thionyl chloride (SOCl₂)

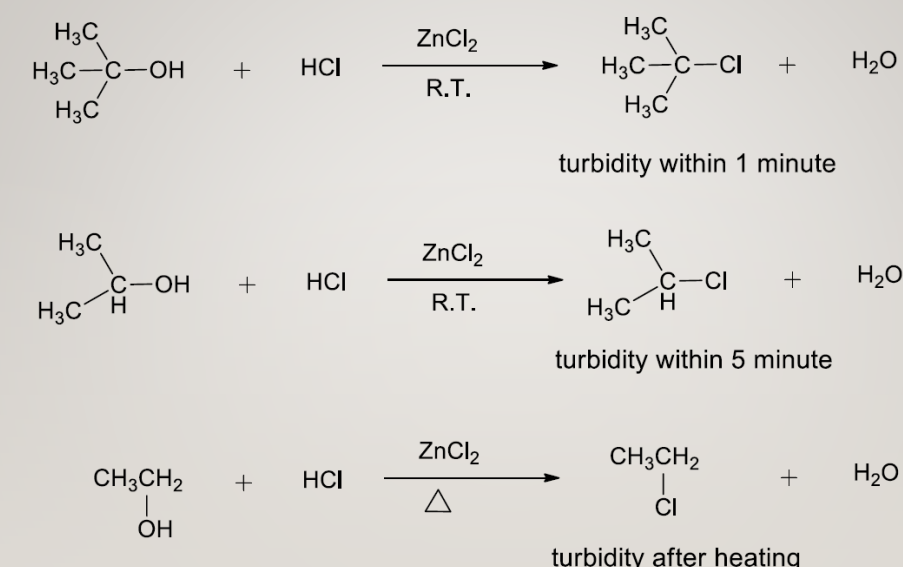
SOCl₂ converts 1° and 2° alcohols to alkyl chlorides without rearrangement.



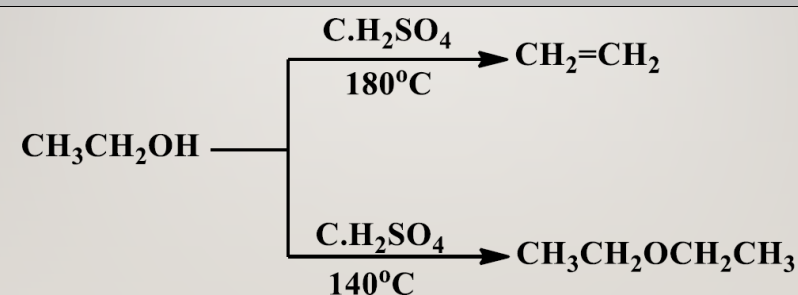
Reaction with ZnCl₂ (Lucas test)

Because the chloride ion is a weaker nucleophile, hydrogen chloride doesn't react with 1° or 2° alcohols unless zinc chloride or some similar Lewis acid is added (Lucas test).

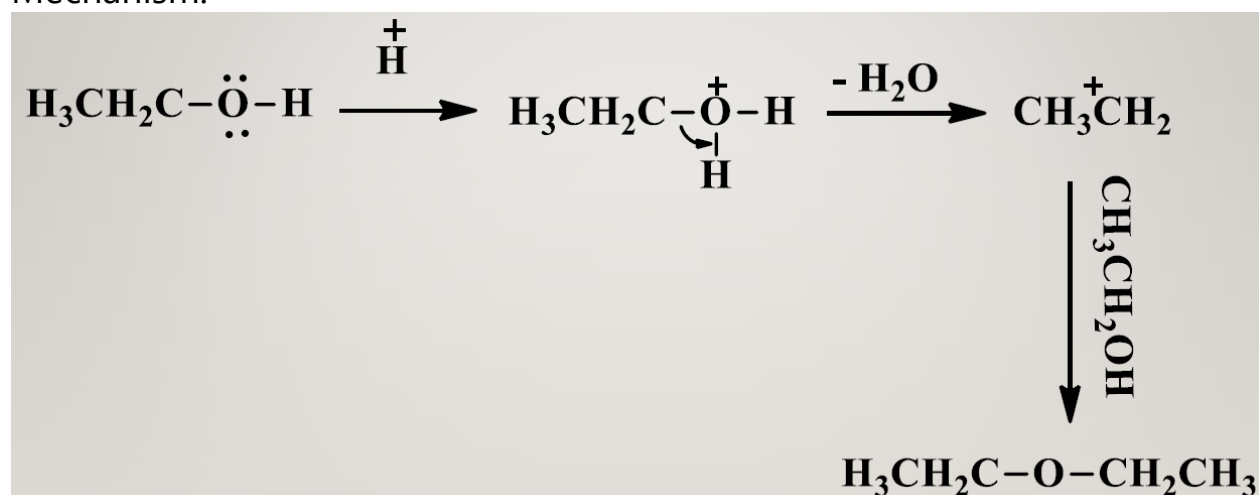
Lucas test can differentiate between primary, secondary and tertiary alcohols as the order of reactivity directed from tertiary to primary alcohols. An equimolar mixture of ZnCl₂ and HCl is Lucas reagent. Tertiary alcohols react immediately with Lucas reagent as evidenced by turbidity owing to the low solubility of the organic chloride in the aqueous mixture. Secondary alcohols react within five minutes. Primary alcohols do not react appreciably with Lucas reagent at room temperature while they react only after heating. Hence, the time taken for turbidity to appear is a measure of the reactivity of the class of alcohol, and this time difference is used to differentiate among the three classes of alcohols.



Reactions of alcohols - 2) Dehydration of alcohol:



Mechanism:





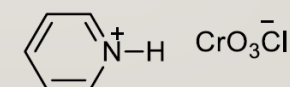
Reactions of alcohols - 3) Oxidation of alcohols :

Non specific oxidizing agents:

As KMnO_4/H^+ or
 $\text{K}_2\text{Cr}_2\text{O}_7/\text{H}^+$ or
 $\text{CrO}_3/\text{CH}_3\text{COOH}$

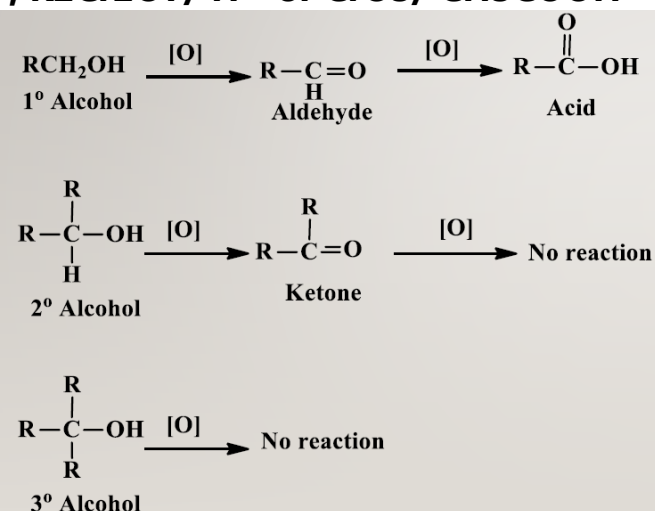
Specific oxidizing agent:

As pyridinium chlorochromate (PCC)



Non specific oxidizing agents:

As KMnO_4/H^+ , $\text{K}_2\text{Cr}_2\text{O}_7/\text{H}^+$ or $\text{CrO}_3/\text{CH}_3\text{COOH}$

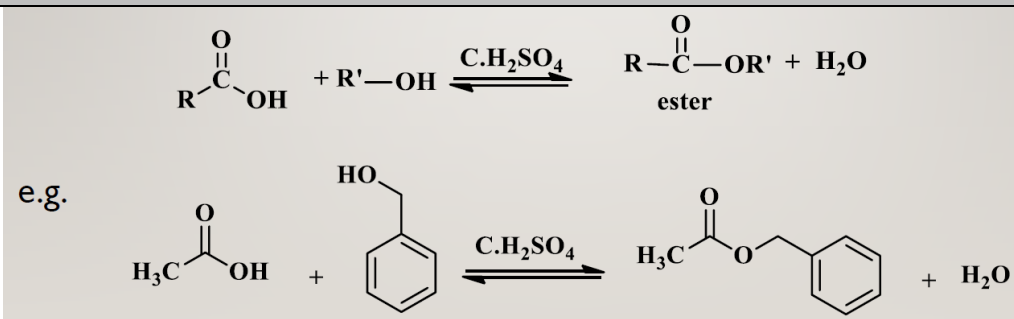


Specific oxidizing agent:

As pyridinium chlorochromate (PCC)

- oxidizes a primary alcohol to an aldehyde and stop at that stage.

Reactions of alcohols - 4) Esterification of alcohol



Mechanism:

